

Customer Journey Map

1. Introduction

A Customer Journey Map illustrates the complete interaction between a user and the OptiCrop system, from identifying a farming problem to receiving a crop recommendation. It helps understand user behavior, expectations, pain points, and opportunities for improving the overall user experience. The primary user of OptiCrop is a farmer seeking scientific guidance for selecting the most suitable crop based on soil and environmental conditions.

2. Objective

The objective of the Customer Journey Map is to visualize the end-to-end experience of users while interacting with the OptiCrop application and to identify areas where the system provides value and improves decision-making.

3. Customer Journey Map

Stage	User Actions	User Goals	Pain Points	OptiCrop Solution
Awareness	Identifies the need for selecting a suitable crop.	Improve agricultural productivity.	Uncertainty about crop selection.	Provides a scientific crop recommendation platform.
Information Gathering	Collects soil nutrient values and environmental information.	Obtain accurate field data.	Limited technical knowledge and expert support.	Requires only essential soil and climate parameters.
Application Usage	Opens the OptiCrop web application and enters the required values.	Receive an accurate crop recommendation.	Difficulty using complex agricultural tools.	Offers a simple and user-friendly Flask web interface.

Prediction	Submits the input values for analysis.	Obtain quick and reliable recommendations.	Concern about prediction accuracy.	Uses a trained Machine Learning model for prediction.
Decision Making	Reviews the recommended crop.	Select the most suitable crop for cultivation.	Fear of making incorrect farming decisions.	Provides data-driven recommendations based on soil and climate conditions.
Outcome	Plans cultivation based on the recommendation.	Increase yield and profitability.	Seasonal and environmental uncertainties.	Supports informed agricultural planning.

4. Customer Experience

Throughout the interaction with OptiCrop, users experience a structured and straightforward workflow. The application minimizes complexity by allowing users to enter seven agricultural parameters and receive an instant crop recommendation. This reduces dependency on traditional decision-making methods and encourages scientific farming practices. The implemented application follows this workflow through a Flask interface that accepts soil and environmental parameters and predicts the most suitable crop.

5. User Expectations

The users expect the system to:

- Provide accurate crop recommendations.
- Generate prediction results quickly.
- Offer a simple and intuitive interface.
- Reduce uncertainty during crop selection.
- Support scientific farming practices.
- Improve agricultural productivity.

6. Opportunities for Improvement

Future enhancements that can further improve the customer journey include:

- Integration with live weather forecasting.
- Fertilizer recommendation module.

- Multi-language support for farmers.
- Mobile application support.
- User login and prediction history.
- Cloud-based deployment for wider accessibility.

7. Benefits of the Customer Journey

The customer journey helps ensure that the system is designed around user needs. It improves usability, increases user confidence, supports informed decision-making, and enhances the overall effectiveness of the OptiCrop application.

8. Conclusion

The Customer Journey Map demonstrates how OptiCrop assists farmers throughout the crop selection process, from identifying the need for guidance to receiving an intelligent recommendation. By focusing on user experience and simplifying interactions, the system promotes efficient, data-driven, and sustainable agricultural practices.